

Ariun Bayasgalan

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EDUCATION

Georgia Institute of Technology

Atlanta, Georgia

Bachelor of Science in Computer Science

May 2028

- o **Concentrations:** Systems Architecture and Info Internetworks
- o **Related Coursework:** Object-Oriented Programming in Java, Python and C++, Data Structures and Algorithms, Systems Programming (Unix/Linux), Software Engineering, Computer Organizations, Computer Networking, Database Management
- o **Leadership & Involvement:** HyTech Racing@GT (autonomous vehicle and embedded systems), WebDev@GT, and IOS Club

TECHNICAL SKILLS

Frontend: React, Javascript, TypeScript, CSS, HTML, Svelte, Tailwind, Figma

Backend: Java, Python, C++, C, Lua, Bash, R, REST APIs, Node.js, Express.js, AWS, PostgreSQL, MySQL, Cypher, MongoDB, Neo4j Graph Databases

Computer Architectures/Frameworks: RISC-V, LC-3, TensorFlow, Pytorch, FastAPI, MERN Stack

Testing/Deployment: CI/CD, Cloudformation, Terraform, OpenTofu, Vercel, Vultr, Azure, AWS EC2, AWS ECS, JUnit, Pytest, Debug Logging

Developer tools: Cursor, Claude code, Codex, Docker, Git, Github, Github Actions, Linux, npm, Agile Methodology, TDD

PROFESSIONAL EXPERIENCE

CACI International, Inc.

May 2026 - Present

Software Engineering, AI/ML Intern

Denver, Colorado

- Founded and led CACI's GraphRAG/Agentic prototyping team and designed 2+ end-to-end delivery frameworks for AI agents and MCP servers, establishing a technical baseline for all future machine learning initiatives.
- Architected GraphRAG MCPs and knowledge graphs using LangChain, Memgraph, Python, Pytest, and H3 geospatial indexing, replacing flat vector information retrieval with spatial and relational context for improved AI output quality and semantic search accuracy.
- Engineered 6+ end-to-end data pipelines that enabled AI systems to process large, previously unsearchable unstructured datasets.
- Tech stack: Python, PyTorch, LangChain, Memgraph, FastAPI, OpenSearch, Docker, Azure, AWS, Temporal, Pydantic, and Pytest.

FIRST Fellowship and MURF Fellowship

Aug 2024 - Dec 2025

Honors Undergraduate Researcher

Golden, Colorado

- Chosen as 1 of ~60 freshmen for a \$1000 FIRST research Fellowship and 1 of ~200 sophomores for a \$1500 MURF research Fellowship.
- Researched dimensional reduction's effect on 3-order topological complexity in Small Language Model (SLM) full weight matrices post-training.
- Investigated 5+ non-von Neumann quantum models for non-classical architectures, enhancing research of quantum computing architecture.
- Tech stack: Python, Topological Data Analysis, R, IBM Quisket, NumPy, PyTorch, Pandas.

Cloud303, Inc.

Dec 2024 - Nov 2025

Software Engineering Intern

Denver, Colorado

- Led the implementation of a secure, internally monitored developer workspace by integrating Coder and AWS EC2, fulfilling a direct CEO initiative.
- Deployed a secure, compliant AWS environment and infrastructure for a European client's US market entry, reducing potential data breach risks by over 50% through strict adherence to European data residency regulations.
- Engineered an EV range predictive analytics ML model via 2+ AWS services (SageMaker, Glue ETL) and CI/CD, delivering data-driven forecasting for a trucking company's zero-emission goals in AWS Infrastructure.
- Designed a secure AWS training and inference pipeline using Pytorch and AWS for auscultation sound classification, improving diagnostic recall.
- Tech stack: Python, TensorFlow, PyTorch, Terraform, CloudFormation, OpenTofu, Docker, AWS, AWS SDK (Boto3).
- AWS: Cloud compute and Containers (EC2, ECS, Lambda, Batch), Data and ML (SageMaker, Glue ETL, DynamoDB, S3, RDS), DevOps and Security (CodePipeline, CodeBuild, API Gateway, EventBridge, GuardDuty, CloudTrail, VPC).

PROGRAMMING PROJECTS

GitTogether | WebDev@GT | MERN Stack, Node.js, Express, React, MongoDB, RESTful API, JWT/Auth, Figma

Jan 2026 - Present

- Launched a web development application for project discovery aimed at 250,000+ Atlanta students using a MERN stack with JWT authentication.

Clean Sight | Python, FastAPI, OpenCV, VectorAI DB, Snowflake, SQLAlchemy, Vultr, SafetyKit, Sphinx, Figma, gRPC

Feb 2026 - Feb 2026

- Architected a full-stack AI platform for hospital infection control, converting surface cleaning sessions into 202-dimensional vectors holding spatial risk embeddings to power real-time infection risk recommendations and clinical decision support.
- Overlaid live 30x20 infection heatmap visualizations by integrating OpenCV onto an MJPEG camera stream, Actian VectorAI DB with gRPC connection and Sphinx for agentic spatial risk reasoning, and SafetyKit for a secure, compliant data pipeline.

NPC Forge | Python, Svelte, Tailwind, AWS (Nova Sonic, Lambda, EC2, DynamoDB, CloudWatch), Agentic AI

Jun 2025 - Jun 2025

- Captured 1 of 4 notable mentions at the Amazon Nova Partner Demo Hackathon.
- Developed an end-to-end agentic web application using Svelte, Tailwind, and 5 AWS services (Nova Sonic, Lambda, EC2, DynamoDB, CloudWatch), yielding an NPC generation platform with persistent player memory and conversational AI.
- Developed a console command to automate Unity-compatible code generation, producing a plug-and-play solution that reduces developer integration time by an estimated 15 hours per project.